

Data Cleaning Change Log

DecodeLabs Industrial Training Kit | Project 1: Data Cleaning & Preparation

Dataset: Dataset_for_Data_Analytics.xlsx | Records: 1,200 | Tool: Python (pandas)

This log documents every check performed on the raw dataset and the corrective action taken for each issue found, per the project's data integrity standard: stakeholders need to know what was changed and why.

Change Summary

Change ID	Description	Impact	Status
CR001	Missing Values: The CouponCode field was missing for 309 out of 1,200 records, about 25.75% of the data. These blanks were filled with the label "No Coupon" instead of using the most common coupon code. A missing value here most likely means the customer simply didn't use a coupon, not that the information was lost. Filling it with a real coupon code would have wrongly implied a discount that was never actually given.	All 1,200 records were preserved, and there are now 0 missing values in this field.	Resolved
CR002	Duplicate Records: The OrderID column, which should contain a unique value for every order, was checked for repeats. All 14 columns together were also checked to catch any rows that were complete copies of another row.	0 duplicate OrderIDs were found, and 0 duplicate rows were found.	Verified
CR003	Date Format: The Date column was already stored correctly as a true date type. Its display format was standardized to ISO 8601 (YYYY-MM-DD) so every date follows the same consistent style.	All 1,200 dates now follow the same format, with 0 incorrectly formatted dates.	Resolved
CR004	Numeric Precision: The UnitPrice column already showed values to 2 decimal places. The TotalPrice column had 29 records with tiny rounding errors caused by how computers store decimal numbers internally, going slightly beyond 2 decimal places. These were rounded to exactly 2 decimal places.	29 records were corrected, leaving 0 precision mismatches.	Resolved
CR005	Text Formatting: The text columns, Product, ShippingAddress, PaymentMethod, OrderStatus, ReferralSource, and CouponCode, were reviewed	No inconsistencies in capitalization or spacing were	Verified

	for inconsistent capitalization, different spellings of the same value, and extra spaces at the start or end of entries.	found across any of the 1,200 records.	
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Verification Gate

Metric	Threshold	Result	Status
Error Rate on Unique Identifiers (OrderID)	0%	0%	Pass
Error Rate on Date Formats	0%	0%	Pass
Missing Values (post-cleaning)	0	0	Pass

Conclusion: This dataset has now been fully cleaned and verified. All missing values were addressed using sound reasoning rather than blindly applying a formula, all records were checked for duplicates with none found, dates were standardized to a single consistent format, and numeric values were corrected to proper precision. Every required check passed the verification gate with zero duplicate identifiers and zero incorrectly formatted dates, meeting the standard set for this project. The cleaned dataset has been saved as `cleaned_dataset.xlsx` and is ready for the next stage of analysis.